J Therm Anal Calorim (2013) 112:245–253
DOI 10.1007/s10973-012-2861-x

Thermodynamic evaluation of the Ga–Y system

M. Idbenali · C. Servant

Phase-Equilibria Special Chapter
© Akadémiai Kiadó, Budapest, Hungary 2013

Abstract The thermodynamic modeling of the gallium–yttrium binary system was carried out with the help of the CALPHAD (CALculation of PHAse Diagram) method. Ga_2Y, GaY, and the three polytypes Ga_3Y_5 have been treated as stoichiometric compounds while a solution model has been used for the description of the liquid phase and the (Ga) and (Y) solid solutions. The excess term of the Gibbs energy of the solution phases was assessed with the recent exponential temperature dependence of the interaction energies by Kaptay (Calphad 28–2:115–24, 1; Calphad 32–2:338–52, 2; Mat Sci Eng A 495:19–26, 3) and compared with the Redlich–Kister (Ind Eng Chem. 4;40:345) polynomial equation results. The calculations based on the thermodynamic modeling are in good agreement with the phase diagram data and experimental thermodynamic values available in the literature.

Keywords Thermodynamic evaluation of Ga–Y system · Calphad method · Redlich–Kister and Kaptay equations · Phase diagram

Introduction

Multicomponent alloys of gallium with rare-earths are of high actuality due to possible application in production of novel magnetic materials [5]. Yttrium containing intermetallic compounds can be used as hard magnetic materials [6]. The knowledge of the thermodynamic properties and phase equilibria in a wide range of concentration and temperature is necessary for the engineering of such new yttrium containing magnetic alloys. The present study is a part of a thermodynamic investigation of Ga-R systems (R = La [7], Ce [8]) which is intended to give a better understanding of the constitutional properties and potential technological applications of these binary alloys. Thermodynamic description is also needed to obtain the thermodynamic factors via the CALPHAD approach for the development of the diffusion database. So far, no thermodynamic description of the Ga–Y system is available in the literature. Our thermodynamic optimization of this binary system is performed based on the experimental data available in the literature (thermodynamic and phase diagram data). In order to avoid the calculation of an unwanted inverted miscibility gap in the liquid phase, thermodynamic constraints were imposed during the optimization: with the Redlich–Kister formalism [4], a positive curvature of the liquidus [9, 10]; and with the more recent Kaptay model [1–3], enthalpy and entropy of same sign.

Experimental data

Phase diagram

The phase diagram of the Ga–Y system was mainly studied by Yatsenko et al. [11] using differential thermal analysis, X-ray diffraction, metallography, and hardness measurements. The nominal purities of the elements (mass%) used to prepare their alloys were Ga (99.999) and Y (99.9). Three gallides Ga_2Y, GaY, and Ga_3Y_5 were identified. The Ga_2Y and GaY compounds melt congruently at 1,623 and 1,658 K, respectively,

M. Idbenali
Centre pédagogique régional, Académie sous massa daraa, Agadir, Morocco
e-mail: idbenalimohamed@yahoo.fr

C. Servant (✉)
Laboratoire de Physicochimie de l'Etat Solide, ICMMO, Université de Paris-Sud, 91405 Orsay Cedex, France
e-mail: coletteservant@orange.fr

while Ga_3Y_5 melts incongruently at 1,568 K. Three eutectic transformations were defined at 1,285–1,288 K (and ~80 at.% Y), at 1,543 K (and ~43 at.% Y), and at 303 K (degenerate eutectic on the Ga-rich side), respectively. The (Ga) and (Y) terminal solid solution domains are very narrow. The liquidus line in the Ga-rich corner of the diagram was determined using the solubility data of Y in liquid Ga [12]. The only difference between the two versions of the experimental phase diagram published by the same authors [11] and [12] is the curvature of this liquidus in the Ga-rich corner at low temperature from 900 to 273 K. The three data indicated on the experimental phase diagram between 700 and 500 °C, on Fig. 2 of the paper by [12] which condition this curvature was previously determined by [13]. Authors [14–16] have also partially contributed to the evaluation of the Ga–Y experimental phase diagram.

Further investigations of the Ga–Y system showed the existence of other gallides previously unknown: Ga_6Y [17] which is an intermetallic superconductor [18], and a Pauli paramagnet [19], Ga_5Y_3, Ga_2Y_3 [20] and the polymorphism of Ga_3Y_5 [21]. Three forms were cited: H_Ga_3Y_5 and M_Ga_3Y_5 which are stable at temperatures above 1,000 and 800 °C, respectively; and L_Ga_3Y_5 stable below 800 °C. Recently, Krachan et al. [22] confirmed only the existence of the three binary compounds Ga_2Y, GaY, and H_Ga_5Y_3. In fact, these authors did not identify the Ga_6Y compound in the Ga–Y binary system but only in the ternary system Ag-Ga–Y in a three phase domain. The Ga–Y phase diagram published by Massalski [23] shows the three phases Ga_2Y, GaY, H_Ga_3Y_5 and an interrupted line at high temperature for Ga_5Y_3. In Table 1 the structural data for the intermetallic compounds of the Ga–Y system are given.

Thermodynamics

The $\Delta_{mix}\overline{H}$ thermodynamic properties of the liquid Ga–Y alloys were measured at 1,125 K up to the yttrium mole fraction x(Y) of 0.22 by [24]. They were also studied using high-temperature calorimetry at 1,750 K by Kanibolotsky et al. [6]. The $\Delta_{mix}\overline{H}$ yttrium partial enthalpy of mixing was measured up to the yttrium mole fraction of 0.6 and the integral enthalpy of mixing was calculated by the Darken's method. The $\Delta\overline{H_{\gamma}^{\infty}}$ yttrium partial enthalpy of mixing at infinite dilution was measured by [24–26] and calculated from Miedema's model in [27]. Merker [25] has determined the enthalpies of formation of the intermetallic compounds Ga_3Y_5, GaY, and Ga_2Y by solution calorimetry. Yamshchikov et al. [26] have established the solubility of yttrium in gallium by molten salt EMF method and reported the Gibbs free energy of formation of Ga_2Y. Later on, Meschel and Kleppa [28] have measured the standard enthalpies of formation of GaY and Ga_2Y by direct synthesis calorimetry. The standard enthalpies of formation of the intermetallic compounds Ga_3Y_5, GaY, and Ga_2Y at 298.15 K were determined by Babu et al. [24] using high-temperature liquid gallium solution calorimetry. These authors also determined the enthalpy of formation of Ga_2Y at 1,125 K by precipitation calorimetry. This enthalpy was derived from the measurements of the $\Delta_{mix}\overline{H}$ integral enthalpies of formation of the Ga–Y liquid alloys when a change of slope of the curve $\Delta_{mix}\overline{H}$ versus x(Y) is noted. Furthermore De Boer et al. [27] calculated the enthalpies of formation of Ga_2Y, Ga–Y, and Ga_3Y_5 by a semi-empirical model. As for the Ga_6Y compound, only its crystal structure and magnetic properties were published in the literature. It has not been mentioned on the two versions of the Ga–Y experimental phase diagram by [11, 12]. Furthermore no experimental data of enthalpy of formation, as well as no invariant reactions were determined for this gallide, so it was not considered in the present assessment of the Ga–Y system.

Thermodynamic models

Pure elements

The Gibbs energy function (298.15 K) for the element i (i = Ga, Y) in the ϕ phase (ϕ = Liquid, ORTHORHOMBIC,

Table 1 Symbols and crystal structures of the solid phases in the Ga–Y system

Phase	Composition Y at.%	Pearson symbol	Space group	Structure designation	Prototype
αGa	0	*oS8*	*Cmca*		αGa
Ga_6Y	14.29	*o2* [17]	*P4/nbm*		Ga_6Pu
Ga_2Y	33.33	*hP3* [34]	*P6/mmm*	*C32*	AlB_2
GaY	50	*oS8* [35]	*Cmcm*		T1I
L_Ga_3Y_5	62.5	*tP32* [32]	*P4/ncc*	–	Ba_5Si_3
M_Ga_3Y_5	62.5			–	M_ Ga_3Y_5
H_Ga_3Y_5	62.5 [33]–61.5 [36]	*hP16* [11, 33]	*P6₃/mcm*	*D88*	Mn_5Si_3
βY	100	*cI2*	*Im* $\overline{3}m$	*A2*	W
αY	100	*hP2*	*P6₃/mcm*	*A3*	Mg

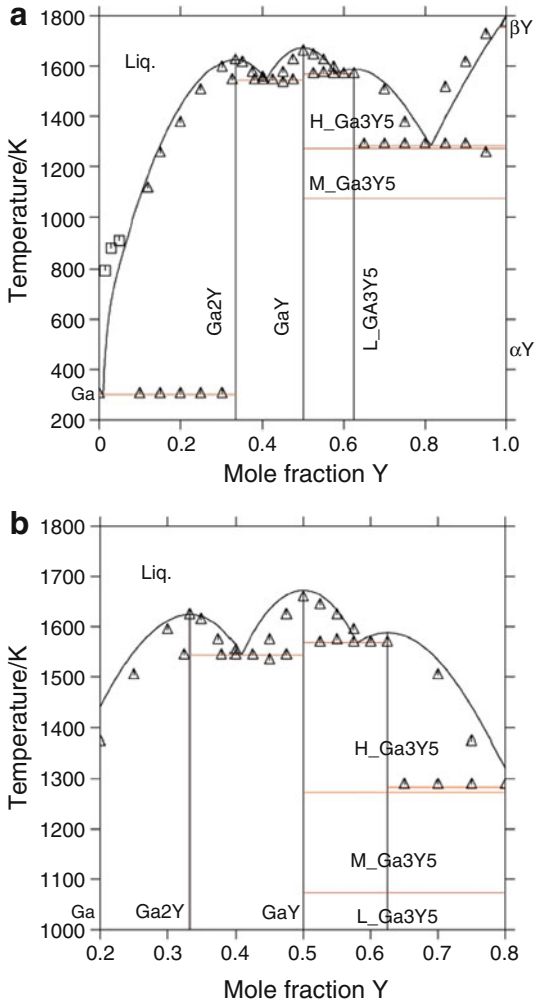

Fig. 1 **a** Calculated Ga–Y phase diagram (Redlich–Kister equation) with the experimental data by [11–12 *open triangle*, 13 *open square*]. **b** Enlarged part of Fig. 1

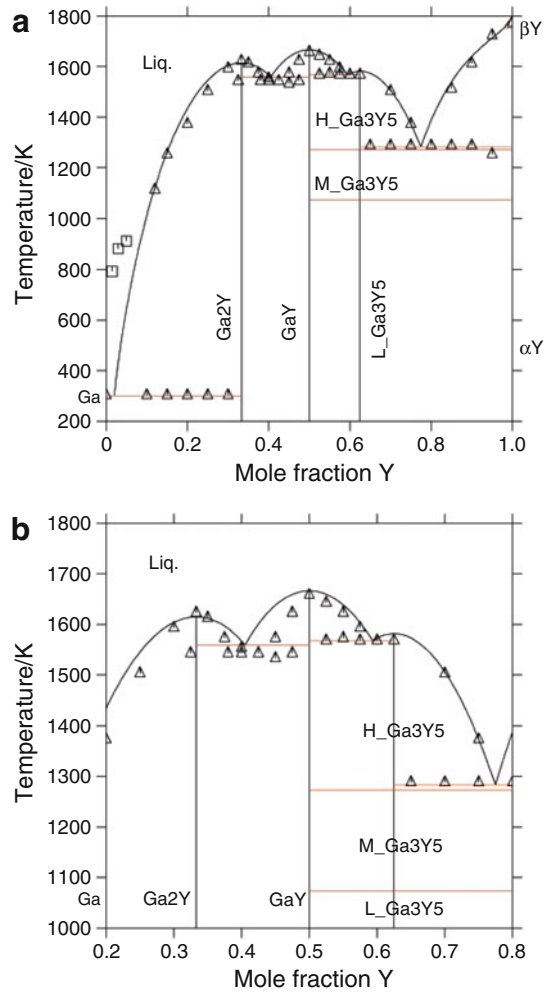

Fig. 2 **a** Calculated Ga–Y phase diagram (Kaptay equation) with the experimental data by [11–12 *open triangle*, 13 *open square*]. **b** Enlarged part of Fig. 2

BCC_A2 or HPC_A3) is described by an equation of the following form:

$$G_i^{\varphi}(T) = a + bT + cT\ln T + dT^2 + eT^3 + fT^7 + gT^{-1} + hT^{-9} \quad (1)$$

In this paper, the Gibbs energy functions are taken from the SGTE compilation of Dinsdale [29]. These functions are valid up to $T = 3{,}700$ and 4,000 K for Yttrium and Gallium, respectively.

Solution phases

The Gibbs energy of one mole of formula unit is expressed as the sum of three terms:

$$G^{\mathrm{liq}} - H^{\mathrm{SER}} = {}^{\mathrm{ref}}G^{\mathrm{liq}} + {}^{\mathrm{id}}G^{\mathrm{liq}} + {}^{\mathrm{ex}}G^{\mathrm{liq}} \quad (2)$$

$$^{\mathrm{ref}}G^{\mathrm{liq}} = x_{\mathrm{Ga}}\left[{}^{0}G^{\mathrm{liq}}_{\mathrm{Ga}} - H^{\mathrm{SER}}_{\mathrm{Ga}}\right] + x_{\mathrm{Y}}\left[{}^{0}G^{\mathrm{liq}}_{\mathrm{Y}} - H^{\mathrm{SER}}_{\mathrm{Y}}\right] \quad (3)$$

where $H_{\mathrm{i}}^{\mathrm{SER}}$ (298.15 K) is the molar enthalpy of the *i* element at 298.15 K in its standard element reference (SER) state, ORTHORHOMBIC for Ga and HCP_A3 for Y.

$$^{\mathrm{id}}G^{\mathrm{liq}} = RT(x_{\mathrm{Ga}}\ln x_{\mathrm{Ga}} + x_{\mathrm{Y}}\ln x_{\mathrm{Y}}) \quad (4)$$

where R is the ideal gas constant; T is the temperature, in Kelvin; x_{Ga} and x_{Y} are the mole fraction of the elements Ga and Y, respectively.

The $^{\mathrm{ex}}G^{\mathrm{liq}}$ energy part in Eq. (2) is given by the Redlich–Kister polynomial [4]

$$^{\mathrm{ex}}G^{\mathrm{liq}} = y_{\mathrm{i}}y_{\mathrm{j}}\sum_{\lambda=0}^{\lambda} {}^{\lambda}L^{\mathrm{Liq}}_{\mathrm{i,j}}(y_{\mathrm{i}} - y_{\mathrm{j}})^{\lambda} \quad (5)$$

y_{i} and y_{j} denotes the site fraction in the sublattice, *i* and *j* the indices which correspond to the two species Gallium and Yttrium and $y_{\mathrm{i}} + y_{\mathrm{j}} = 1$.

Table 2 The optimized thermodynamic parameters of the Ga–Y system (Redlich–Kister equation)

Phase	Thermodynamic model	Parameters/J mol^{-1} at.$^{-1}$ and J mol^{-1} at.$^{-1}$ K^{-1}
Liquid	(Ga, Y)	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Ga:Y}} = -274223.227 + 80.5535975\,T - 0.015405174\,T^2$
		${}^{1}L^{\mathrm{Liq}}_{\mathrm{Ga:Y}} = -90947.7118 + 75.493407\,T$
		${}^{2}L^{\mathrm{Liq}}_{\mathrm{Ga:Y}} = 125165.064 - 48.3473188\,T$
BCC_A2	$(\mathrm{Ga, Y})_1(\mathrm{Va})_3$	No excess term optimized
HCP_A3	$(\mathrm{Ga, Y})_1(\mathrm{Va})_{0.5}$	No excess term optimized
ORTHO_Ga	(Ga)	No excess term optimized
Ga_2Y	$(\mathrm{Ga})_{0.667}$:$(\mathrm{Y})_{0.333}$	$G^{\mathrm{Ga2Y}}_{\mathrm{Ga:Y}} - 0.667\,{}^{298}H^{\mathrm{ORTHORHOMBIC_Ga}}_{\mathrm{Ga}} - 0.333\,{}^{298}H^{\mathrm{HCP}}_{\mathrm{Y}} =$
		$0.3333 * \mathrm{GHSERY} + 0.6667 * \mathrm{GHSERGA} - 73491.8966 + 7.09297275\,T$
GaY	$(\mathrm{Ga})_{0.5}$:$(\mathrm{Y})_{0.5}$	$G^{\mathrm{GaY}}_{\mathrm{Ga:Y}} - 0.5\,{}^{298}H^{\mathrm{ORTHORHOMBIC_Ga}}_{\mathrm{Ga}} - 0.5\,{}^{298}H^{\mathrm{HCP}}_{\mathrm{Y}} =$
		$0.5 * \mathrm{GHSERY} + 0.5 * \mathrm{GHSERGA} - 75200.4417 + 4.55880737\,T$
$L_Ga_3Y_5$		$G^{\mathrm{L_Ga3Y5}}_{\mathrm{Ga:Y}} - 0.375\,{}^{298}H^{\mathrm{ORTHORHOMBIC_Ga}}_{\mathrm{Ga}} - 0.625\,{}^{298}H^{\mathrm{HCP}}_{\mathrm{Y}} =$
		$-64200.00 + 40.03796T - 14.2602116T\mathrm{Ln}T$
$M_Ga_3Y_5$	$(\mathrm{Ga})_{0.375}$:$(\mathrm{Y})_{0.625}$	$G^{\mathrm{M_Ga3Y5}}_{\mathrm{Ga:Y}} - 0.375\,{}^{298}H^{\mathrm{ORTHORHOMBIC_Ga}}_{\mathrm{Ga}} - 0.625\,{}^{298}H^{\mathrm{HCP}}_{\mathrm{Y}} =$
		$-64100.00 + 39.9448T - 14.2602116T\mathrm{Ln}\,T$
$H_Ga_3Y_5$		$G^{\mathrm{H_Ga3Y5}}_{\mathrm{Ga:Y}} - 0.375\,{}^{298}H^{\mathrm{ORTHORHOMBIC_Ga}}_{\mathrm{Ga}} - 0.625\,{}^{298}H^{\mathrm{HCP}}_{\mathrm{Y}} =$
		$-63996.5604 + 39.8635256T - 14.2602116T\ln T$

Va vacancy

Table 3 The optimized thermodynamic parameters of the Ga–Y system (Kaptay equation)

Phase	Thermodynamic model	Parameters/J mol^{-1} at.$^{-1}$ and J mol^{-1} at.$^{-1}$ K^{-1}
Liquid	(Ga, Y)	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Ga:Y}} = -274223.227 \exp\,(-4.41668972\mathrm{E}-04\,T)$
		${}^{1}L^{\mathrm{Liq}}_{\mathrm{Ga:Y}} = -88560.0471 \exp\,(-1.14631637\mathrm{E}-03\,T)$
		${}^{2}L^{\mathrm{Liq}}_{\mathrm{Ga:Y}} = 122842.391 \exp\,(-8.11353252\mathrm{E}-04\,T)$
BCC_A2	$(\mathrm{Ga, Y})_1(\mathrm{Va})_3$	No excess term optimized
HCP_A3	$(\mathrm{Ga, Y})_1(\mathrm{Va})_{0.5}$	No excess term optimized
ORTHO_Ga	(Ga)	No excess term optimized
Ga_2Y	$(\mathrm{Ga})_{0.667}$:$(\mathrm{Y})_{0.333}$	$G^{\mathrm{Ga2Y}}_{\mathrm{Ga:Y}} - 0.667\,{}^{298}H^{\mathrm{ORTHORHOMBIC_Ga}}_{\mathrm{Ga}} - 0.333\,{}^{298}H^{\mathrm{HCP}}_{\mathrm{Y}} =$
		$0.3333 * \mathrm{GHSERY} + 0.6667 * \mathrm{GHSERGA} - 73491.8966 + 11.7344572\,T$
GaY	$(\mathrm{Ga})_{0.5}$:$(\mathrm{Y})_{0.5}$	$G^{\mathrm{GaY}}_{\mathrm{Ga:Y}} - 0.5\,{}^{298}H^{\mathrm{ORTHORHOMBIC_Ga}}_{\mathrm{Ga}} - 0.5\,{}^{298}H^{\mathrm{HCP}}_{\mathrm{Y}} =$
		$0.5 * \mathrm{GHSERY} + 0.5 * \mathrm{GHSERGA} - 75190.4404 + 12.3226546\,T$
$L_Ga_3Y_5$		$G^{\mathrm{L_Ga3Y5}}_{\mathrm{Ga:Y}} - 0.375\,{}^{298}H^{\mathrm{ORTHORHOMBIC_Ga}}_{\mathrm{Ga}} - 0.625\,{}^{298}H^{\mathrm{HCP}}_{\mathrm{Y}} =$
		$-64200 + 142.00565\,T - 26.9059083\,T\ln T$
M_Ga_3Y	$(\mathrm{Ga})_{0.375}$:$(\mathrm{Y})_{0.625}$	$G^{\mathrm{M_Ga3Y5}}_{\mathrm{Ga:Y}} - 0.375\,{}^{298}H^{\mathrm{ORTHORHOMBIC_Ga}}_{\mathrm{Ga}} - 0.625\,{}^{298}H^{\mathrm{HCP}}_{\mathrm{Y}} =$
		$-64100 + 141.912450\,T - 26.9059083\,T\ln T$
$H_Ga_3Y_5$		$G^{\mathrm{H_Ga3Y5}}_{\mathrm{Ga:Y}} - 0.375\,{}^{298}H^{\mathrm{ORTHORHOMBIC_Ga}}_{\mathrm{Ga}} - 0.625\,{}^{298}H^{\mathrm{HCP}}_{\mathrm{Y}} =$
		$-64000 + 141.833876\,T - 26.9059083\,T\ln T$

Va vacancy

The binary interaction parameters of the ${}^{\lambda}L^{\mathrm{Liq}}_{\mathrm{i,j}}$ type, assessed in the present work, were temperature dependent, as follows:

First from [4] : ${}^{\lambda}L^{\mathrm{Liq}}_{\mathrm{i,j}} = a_{\lambda} + b_{\lambda}T$ (6)

with the excess entropy $s = -b_{\lambda}$ (knowing for example that for lambda = 0, excess entropy versus composition is calculated through this coefficient—b and is then $y_i{*}y_j{*}b$).

Secondly from [1–3] : ${}^{\lambda}L^{\mathrm{Liq}}_{\mathrm{i,j}} = h^{\lambda}_{\mathrm{oi}} * \exp\left(-\frac{T}{\tau^{\lambda}_{\mathrm{oi}}}\right)$ (7)

Table 4 Invariant reactions in the Ga–Y System (Redlich–Kister and Kaptay equations)

Invariant reactions		[11]		This work Kister		This work
	T/K	*x*(Y)/at	(Redlich equation) (*T*/K)	*x*(Y)/at	(Kaptay equation) *T*/K	*x*(Y)/at
(Ga) ⇔ Liq + Ga_2Y	303	Liq 0.0001	302	Liq 0.0057	300.7	Liq 0.0235
Liq ⇔ Ga_2Y	1,623	0.3333	1,624	0.332	1,615	0.3367
Liq ⇔ Ga_2Y + GaY	1,543	Liq 0.43	1,544	Liq 0.46	1,559	Liq 0.4053
Liq ⇔ GaY	1,658	0.50	1,672	0.50	1,666	0.50
H_Ga_3Y_5 ⇔ Liq + GaY	1,568	Liq 0.625	1,568	Liq 0.58	1,568	Liq 0.5937
Liq ⇔ H_Ga_3Y_5 + α(Y)	1,286–1,288	Liq 0.8	1,283	Liq 0.815	1,283	Liq 0.7747
H_Ga_3Y_5 ⇔ M_Ga_3Y_5 + α(Y)	1,273	0.625 0.615	1,273	0.625	1,273	0.625
M_Ga_3Y_5 ⇔ L_Ga_3Y_5 + α(Y)	1,073	0.625	1,073	0.625	1,073	0.625
α(Y) ⇔ β(Y) + Liq	1,751	Liq 0.963	1,751	Liq 0.982	1,752	Liq 0.9786

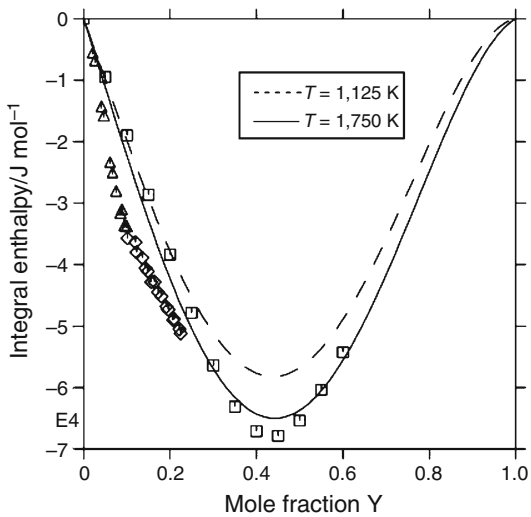

Fig. 3 Calculated integral enthalpies of formation of the liquid phase (Redlich–Kister equation) versus *x*(Y) with the experimental data by [24]: *open triangle* (one phase), *open diamond* (two phases) T = 1,125 K and [2]: *open square* $T = 1{,}750$ K, *continuous line* and *dashed line* this work

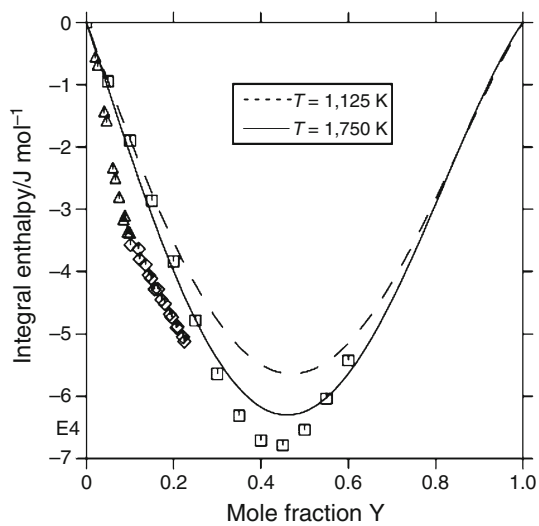

Fig. 4 Calculated integral enthalpies of formation of the liquid phase (Kaptay equation) versus *x*(Y) with the experimental data by [24]: *open triangle* (one phase), *open diamond* (two phases) $T = 1{,}125$ K and [2]: *open square* $T = 1{,}750$ K *continuous line* and *dashed line* this work

with the h^{λ}_{oi} enthalpy part (J mol^{-1}) of ${}^{\lambda}L^{Liq}_{i,j}$ at $T = 0$ K; the τ^{λ}_{oi} must be always positive (the unit of τ is K). By identifying h^{λ}_{oi} with a_{λ} and τ^{λ}_{oi} with a_{λ}/b_{λ}, (Eq. 7) becomes (Eq. 8) with the excess entropy $s = b_{\lambda}$.

$$ {}^{\lambda}L^{Liq}_{i,j} = a_{\lambda} * \exp((-b_{\lambda}/a_{\lambda}) * T) \quad (8) $$

a_{λ} and b_{λ} are the coefficients to be optimized.

In the (Ga) and (Y) solid solutions, the respective solubility of Y and Ga is very low, consequently no excess term of the Gibbs energy was optimized for these terminal solid solutions.

Stoichiometric compounds

The L_Ga_3Y_5, M_Ga_3Y_5, H_Ga_3Y_5, GaY, and Ga_2Y gallides were considered as stoichiometric compounds. Except for the three Ga_3Y_5 polytypes, their Gibbs energy noted as ${}^{0}G_{ApBq}$ was expressed as follows:

$$ {}^{0}G_{ApBq} = \frac{p}{p+q}{}^{0}G_{A} + \frac{q}{p+q}{}^{0}G_{B} + a + bT \quad (9) $$

where ${}^{0}G_{A}$ and ${}^{0}G_{B}$ are the Gibbs energy of the pure elements Gallium and Yttrium, respectively; *a* and *b* are

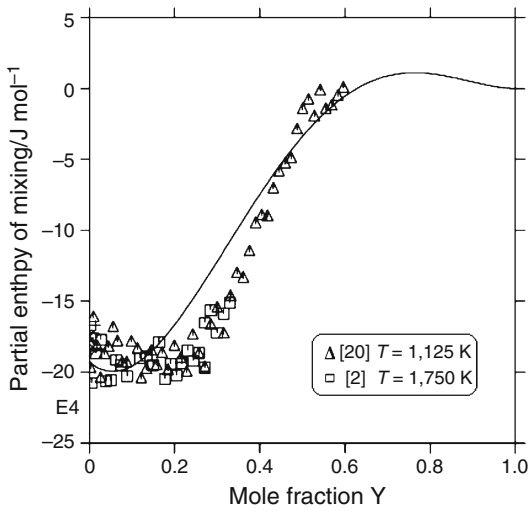

Fig. 5 Calculated partial enthalpies of mixing of the liquid phase (R-K equation) versus x(Y) with the experimental data by [20] *open triangle*: $T = 1{,}125$ K and [2] *open square*: $T = 1{,}750$ K, *continuous line* this work

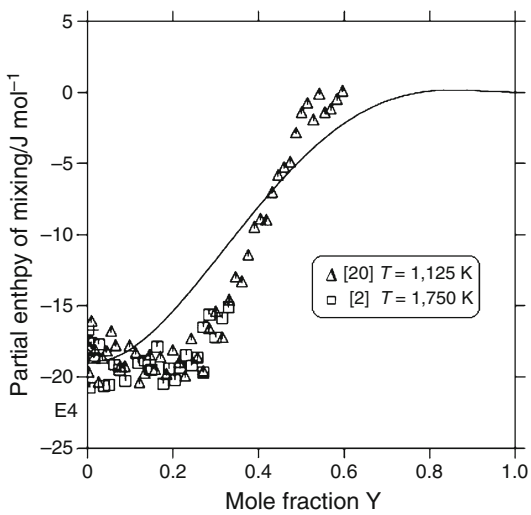

Fig. 6 Calculated partial enthalpies of mixing of the liquid phase (Kaptay equation) versus x(Y) with the experimental data by [20] *open triangle* $T = 1{,}125$ K and [2] *open square* $T = 1{,}750$ K, *continuous line* this work

parameters to be determined,. For the polytypes, the Gibbs energy was expressed as follows:

$$^{0}G_{\mathrm{ApBq}} = +a + bT + cT\ln T \tag{10}$$

Indeed, a self-consistent set of thermodynamic parameters was better obtained with Eq. 10 than Eq. 9. In fact, as a c heat capacity was introduced in $^{0}G_{\mathrm{ApBq}}$, it was not necessary to use the $\mathrm{GHSER_{GA}}$ and $\mathrm{GHSER_{Y}}$ functions [29] which also contain a c term.

Results and discussions

The evaluation of the thermodynamic parameters was performed by the optimization module PARROT within the Thermo-Calc software [30], which works by minimizing the square sum of the differences between measured and calculated values. The liquid was treated as a substitutional solution. Only the three intermetallic phases (Ga_2Y, GaY, and Ga_3Y_5) were taken into account and treated as stoichiometric compounds. The different polytypes of Ga_3Y_5 were considered using Eq. 10. The reliable experimental data marked in Sect. 2 were employed in the optimization. The thermodynamic parameters of Eqs. (9, 10) were adjusted according to the enthalpies of formation measured by [24–26, 28] and predicted by [27]. The compositions of the phases and the temperatures involved in the invariant reactions mainly based on the phase diagram of the Ga–Y system evaluated by [11, 12] were used and all the thermodynamic parameters were optimized simultaneously. In addition, in order to avoid the formation of an unwanted inverted miscibility gap of the liquid solution phase during the calculation of the Ga–Y phase diagram, constraints have been imposed during the optimization procedure [9, 10] when using the two formalisms: Redlich–Kister [4] and Kaptay [1–3]. In the first procedure, a positive curvature of the liquidus by imposing the additional constraint $d^2G/dx^2 > 0$ in the atomic composition range $0 < x(\mathrm{Y}) < 1$ and every 50° from the liquidus temperature up to 6,000 K was, therefore, optimized. The calculated Ga–Y phase diagram from the optimized parameters is shown in Fig. 1a, b. In the second formalism (enthalpy and entropy of the excess term of the liquid phase imperatively of same sign), the optimized parameters led to the calculated phase diagram shown in Fig 2a, b. The two calculated diagrams are similar. The liquidus line in the Ga-rich corner below 900 K was optimized between the two versions of the experimental diagram by [11–13]. In the case of the optimization with the Kaptay equation [1–3], the liquidus slope goes down more continuously from the congruent melting of Ga_2Y toward $T = 273$ K than in the case of the optimization with the Redlich–Kister equation [4] for which a more abrupt slope is observed between 700 and 300 K. While in [11, 12], the invariant peritectic reaction $\mathrm{Liq} + \mathrm{GaY} \leftrightarrow \mathrm{H_Ga_3Y_5}$ is somewhat degenerate, the liquid and the $\mathrm{H_Ga_3Y_5}$ compound have the same composition, our optimizations led to a congruent point for $\mathrm{H_Ga_3Y_5}$ at 1,588 K (R-K)/1,582 K (Kaptay) and to a eutectic reaction $\mathrm{Liq} \leftrightarrow \mathrm{GaY} + \mathrm{H_Ga_3Y_5}$ at 1,568 K, Figs. 1a and Fig. 2a. The difference of composition between the liquid and the $\mathrm{H_Ga_3Y_5}$ compound is weak (~ 0.03 at Y).

It must be mentioned that in the process of optimization with the Redlich–Kister formalism, two, three, or even four parameters of Eq. (6) were used, respectively, in order to get a good description for the liquid phase. In fact, a T^2

Table 5 Enthalpies of formation of the intermetallic compounds

Compounds	$-\Delta_f^0 H_{298.15}$ kJ mol^{-1} at.$^{-1}$ (exp.) Literature	Reference	This work Redlich–Kister eq. [R-K] Kaptay eq. [K]	Method	$-\Delta_f^0 H_{298.15}$ kJ mol^{-1} at.$^{-1}$ (pred.) [27]
Ga_6Y	35.0		Interpolated		
Ga_2Y	61.8 ± 2.8	[24]	73.47 [R-K]	Precipitation	65.0
			73.49 [K]	Cal. 1,125 K	
	64.4 ± 2.5	[24]		Ga Soln. Calo	
				1,099 K.	
	68.1 ± 2.3	[28]		Dir. Syn. Calo.	
				1,373 ± 2 K	
	69 .0	[25]		Ga Soln. Calo	
				1,176 K	
	72.1 ± 1.3 [28]	[26]		Emf	
	68.4 ± 1.3 [26]			(854–1,150 K)	
	74.6 ± 1.5 [28]	[26]		Emf	
	70.9 ± 1.5 [26]			(650–854 K)	
GaY	68.9 ± 2.8	[24]	75.20 [R-K]	Ga Soln. Calo	73.0
			75.19 [K]	1,095 K	
	71.4 ± 1.7	[28]		Dir. Syn. Calo.	
				1,373 K	
	71	[25]		Ga Soln. Calo	
				1,176 K	
Ga_3Y_5	57.0	[25]	63.99 [R-K]	Ga Soln. Calo	61.0
			64.00 [K]	1,176 K	
	59.6 ± 3.6	[24]		Ga Soln. Calo	
				1,103 K	

term of the excess parameters of degree 0 was necessary to be optimized to give account of the respective mixing enthalpies of formation of the liquid phase measured at 1,125 K by [24] and at 1,750 K by [5] (Table 2).

By comparing the thermodynamic parameters obtained by the optimizations made with the two equations of the temperature dependence of the excess term of the Gibbs energy of the liquid phase, it will be noted that the a_λ (J mol^{-1} at.$^{-1}$) enthalpy part of the three ${}^{\lambda}L_{i,j}^{Liq}$ excess terms of degree 0 [−274223.22 (R-K) or −274223.22 (Kaptay)], degree 1 [−90947.71 (R-K) or −88560.04 (Kaptay)], and degree 2 [125165.06 (R-K) or 122842.39 (Kaptay)] are very similar (Tables 2, 3). On the contrary, the entropy part in J mol^{-1} at.$^{-1}$ K^{-1} ($s = -b_\lambda$ for R-K) and ($s = +b_\lambda$ calculated from b_λ/a_λ for Kaptay equation) is generally somewhat different: (−80.55 or −121.11), (−75.49 or −101.51), and (+48.34 or +99.66) due to a linear temperature dependence instead of an exponential dependence. The invariant equilibria of the Ga–Y system are listed in Table 4. As shown in this table, a good agreement is obtained between the calculations and the experiments. It will be noted that all the digits of the thermodynamic parameters must be used to calculate the temperatures of the invariant reactions and the compositions of the phases involved, in particular when the polytypes are concerned. The $\Delta_{mix}\overline{H}$ calculated integral enthalpies of mixing of the liquid Ga–Y alloys referring to liquid components at $T = 1{,}125$ K and at $T = 1{,}750$ K are plotted in Figs. 3 and 4 with the experimental data. A better behavior of the curve is observed with the formalism of Kaptay in the Y-rich part, around $x(Y) = 0.9$–1 at. The experimental and calculated enthalpies of mixing of the liquid phase are dependent on temperature. They are more negative for the lower temperature. The higher difference (around 9 kJ mol^{-1} at.$^{-1}$) between the curves calculated at 1,125 and 1,750 K is reached for $x(Y) = 0.44$ at. Figures 5 and 6 show the very similar $\Delta_{mix}\overline{H}$ calculated partial mixing enthalpies of liquid Y in liquid Ga compared with the experimental data of [6, 25]. As a function of $x(Y)$ the curve is first decreasing, then monotonously increasing up to a plateau. However, a discrepancy exists between our calculated results and the experimental ones. This is due to the weak level of relevance attributed to this information in the optimization in order to keep consistency with other selected data. The calculated $\Delta_{mix}\overline{H_\gamma^\infty}$ value of infinite dilution of

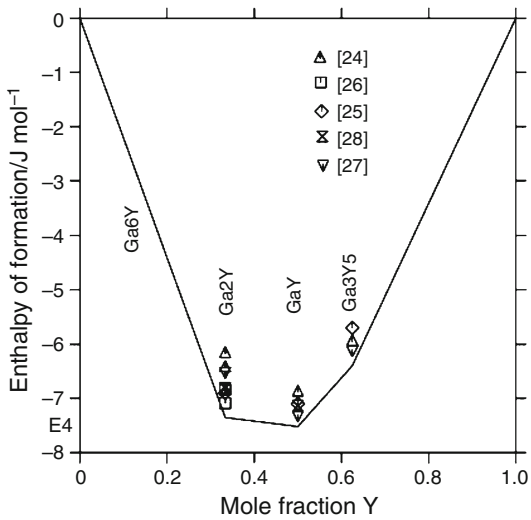

Fig. 7 Calculated enthalpies of formation of the intermetallic compounds at 298.15 K versus x(Y), (R-K and Kaptay equations) with the experimental data (*continuous line* this work)

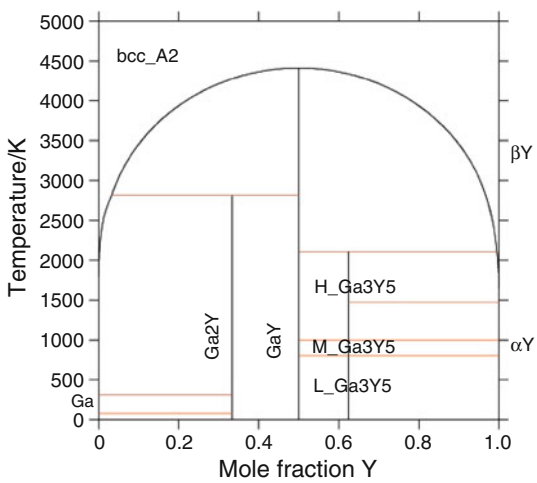

Fig. 8 Calculated Ga–Y phase diagram (Redlich–Kister equation) when the liquid phase is suspended

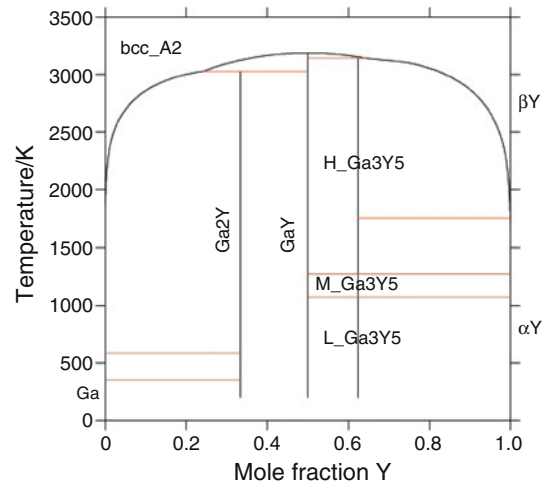

Fig. 9 Calculated Ga–Y phase diagram (Kaptay equation) when the liquid phase is suspended

liquid Y in liquid Ga (R-K equation) at 1,750 K (-181 kJ mol^{-1} at.$^{-1}$) is in good agreement with the data (-181.63 ± 7.93 kJ mol^{-1} at.$^{-1}$) by [6] or (-184.4 ± 5.3 kJ mol^{-1} at.$^{-1}$) by [24] at 1,099 K or the predicted value (-180.25 kJ mol^{-1} at.$^{-1}$) by [27] at 14 K (Table 5).

The calculated and the available experimental values of the enthalpies of formation of the intermetallic compounds are listed in Table 4 and presented in Fig. 7. The enthalpy part of formation is identical in the two formalisms but the entropy part is slightly higher in the Kaptay procedure; consequently these compounds are less stable at high temperature than in the Redlich–Kister formalism, this explains the difference between the two graphs of phase diagram calculated in Figs. 8 and 9 when the liquid phase is suspended. For both optimizations, the enthalpy and entropy parts of formation of the polytypes are very near. The enthalpy part is decreasing of about 200 J mol^{-1} at.$^{-1}$ and the entropy part is decreasing of about 0.17 J mol^{-1} at.$^{-1}$ K^{-1} from L_Ga_3Y_5 to M_Ga_3Y_5 then to H_Ga_3Y_5. A satisfactory agreement is noted. From Fig. 7, it is possible to interpolate a value of the enthalpy of formation of the hypothetical Ga_6Y gallide, around -35 kJ mol^{-1} at.$^{-1}$.

As mentioned in [31], in order to check that the optimized thermodynamic parameters of the intermetallic compounds are satisfactory, we verified that when the liquid phase is not considered during the calculation of the phase diagram, the stoichiometric phases disappear at high temperatures; the terminal solid solutions and a two-phase domain existing between them is generally calculated. But, in our case, it will be noted that at high temperature, only the (Ga–Y) BCC_A2 solid solution is calculated on the whole Yttrium composition range and not the (Ga) in the Ga-rich region. This is due to the power series in terms of temperature for the Gallium element in the BCC_A2 state which becomes metastable at higher temperatures [29] compared with the ORTHORHOMBIC_GA state.

Conclusions

A self-consistent thermodynamic description of the Ga–Y system—by comparing to describe the excess part of the Gibbs energy the two formalisms of Redlich–Kister (with the $d^2G/dx^2 > 0$ additional constraint) and Kaptay (with the constraint, enthalpy and entropy of same sign) to avoid the formation of a miscibility gap in the solution phases—was obtained based on available experimental information

in the literature (thermodynamic properties and phase diagram data). Most of the experimental thermodynamic and phase equilibrium data are well reproduced. The optimizations made with the two formalisms are equivalent. The calculated phase diagram and thermodynamic properties of the Ga–Y binary system can be used to further understand the thermodynamics of other ternary systems *-Ga–Y such as Ga-Ge–Y.

References

1. Kaptay G. A new equation for the temperature dependence of the excess Gibbs energy of the solution phase. Calphad. 2004; 28–2:115–24.
2. Kaptay G. A Calphad-compatible method to calculate liquid/liquid interfacial energies in immiscible metallic system. Calphad. 2008;32–2:338–52.
3. Kaptay G. A unified model for the cohesive enthalpy, critical temperature, surface tension and volume thermal expansion coefficient of liquid metals of bcc, fcc and hcp crystals. Mat Sci Eng A. 2008;495:19–26.
4. Redlich O, Kister A. Algebraic representation of thermodynamic properties and the classification of solutions. Ind Eng Chem. 1948;40:345.
5. Kanibolotsky DS, Bieloborodova OA, Lisnyak VV. Calorimetric study of enthalpies of mixing in liquid gallium–germanium–yttrium alloys. Thermochim Acta. 2005;438:51.
6. Kanibolotsky DS, Bieloborodova OA, Lisnyak VV. Examination of enthalpies of mixing in liquid gallium–yttrium alloys by high temperature calorimetry. Thermochim Acta. 2005;437:62–6.
7. Idbenali M, Servant C. Communication presented at JEEP2011 Saint-Avold, France, published by EDP Science (2011).
8. [8] Idbenali M., Servant C. Communication presented at Euromat 2011 Montpellier, France.
9. Lupis CH. Chemical thermodynamics of metals. North-Holland: Prentice-Hall Inc.; 1983.
10. Kumar KC, Wollants P. Some guides for thermodynamic optimisation of phase diagrams. J Alloys Comp. 2001;320–2:189–98.
11. Yatsenko SP, Semyannikov AA, Semenov BG, Chuntonov KA. Phase diagrams of rare earth metals with gallium. J Less-Common Met. 1979;64:185–99.
12. Yatsenko SP. Gallii Vzaimodeistvie s Metallami. Gallium Interaction with Metals. Moscow: Nauka; 1974.
13. Semenov BG. Thesis, Nauk. SSSR, UNC, Sverdlovsk 1977.
14. Iandelli AG. Phase diagram. Chim Ital. 1949;79:70.
15. Buschow KHJ, Hoogenhof WW. Magnetic properties and phase relationships of gadolium–galium compounds. J Less-Common Met. 1976;45–2:309–13.
16. Yatsenko SP, Anikin Yu A. Izv. Akad. Nauk. SSSR, Met. 1970;4:162.
17. Gryniv OI, Griniv IA, Gryn YN, Yarmolyuk YP Dop. Akad. Nauk Ukr. RSR, B: geologichni, khimichni ta biologichni Nauk. 1985;5:42.
18. Kadomatsu H, Sakurai J, Sugamoto J, Fujiwara H. Pressure-induced superconductivity in $TbGa_6$ and $DyGa_6$. Solid State Commun. 1992;82:111–4.
19. Tagawa Y, Sakurai J, Komura Y, Ishimasa T. Magnetic susceptibility and electrical resistivity of RGa6 (R = rare earth metals). J Less-Common Met. 1986;119:269.
20. Gryn' YN, Gladyshewsky RE. Gallides. Metallurgia. Moscow, 1989 (in Russian).
21. Markiv V.Ya. Belyavina N.M, Dop. Akad. Nauk Ukr., Rep. Nat. Acad. Sci. Ukr., 1992;10:142.
22. Krachan T, Stel'makhovych B, Kuz'ma Yu. The Y-Ag-Ga system. J Alloys Comp. 2005;386:147–50.
23. Massalski TB (ed.) Binary alloy phase diagrams, 2nd ed. Materials Park: ASM International; 1990. pp. 1874–1875.
24. Babu R, Magarajan K, Venugopal V. Standard enthalpies of formation of yttrium gallides by high temperature reaction calorimetry. J Alloys Comp. 2000;311:200–6.
25. Merker P. Enthalpies of formation of some Ga–Y intermetallic compounds. J Less-Common Met. 1991;69:L23–4.
26. Yamshchikov LF, Lebedev VA, Nichkov IF, Raspopin SP, Kokoulin OK. Russ J Phys Chem. 1979;53(5):657.
27. de Boer FR, Boom R, Mattens WC, Miedema AR, Niessen AK. Cohesion in metals, transition metals alloys. Cohesion and structure, vol. 1. Amsterdam: North-Holland; 1988.
28. Meschel SV, Kleppa OJ. Standard enthalpies of formation of some 4d transition metal gallides by high temperature direct synthesis calorimetry. J Alloys Comp. 2000;297:162–7.
29. Dinsdale AT. SGTE data for pure elements. Calphad. 1991;15: 317–425.
30. Sundman B, Janson B, Andersson J-O. The Thermo-Calc databank system. Calphad. 1985;9:153–90.
31. Chen SL, Daniel S, Zhang Z, Chang YA, Oates WA, Schmid Fetzer RJ. On the calculation of multicomponent stable phase diagrams. J Phase Equilib. 2001;22:373.
32. Zhao J, Corbett JD. R_5Ga_3 compounds of selected rare earth metals R: structures and properties. J Alloys Comp. 1994; 210:1–7.
33. Shob O, Parthé E. Sc_5 and Y_5Ga_3 with $D8_8$ structure. H Acta Cryst. 1964;17:1335–6.
34. Haszko SE. Rare-earth gallium compounds having the Aluminium-Boride structure. Trans Am Inst Min. 1961;221:201–4.
35. Dwight AE, Downey JW, Conner RA. Equiatomic compounds of Y and the lanthanides elements with Ga. Acta Cryst. 1967;23: 860–2.
36. Markiv VYa, Belyavina NM, Speka MV. The isothermal section of the phase diagram of the ternary system Y–Ge–Ga at 800°C in the region 33.3 to 100 at.% Y. J. Alloys Compd. 1999;285: 167–71.